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Patterns

Q1 Print an nxn star square
Take a dynamic input from the user and print an nxn square
Example:
If n = 5

```
* * * * *
* * * * *
* * * * *
* * * * *
```

```
import java.util.*;
class StarPattern {
    public static void main (String[]args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 1; i<=n; i++) {
            for (int j=1; j<=n; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Dry Run
n = 5
i = 1
condition check
↓ true
kaam ⟹ j = ~~1 2 3 4 5 6~~ condition false
↓
inc/dec

condition check
↓ true
kaam ⟹ * * * * *
↓
inc/dec

जर आपल्या User kadun input ghyayche hote जर आपला User ne jr 4 input di jr 4x4 square pattern tyar honar mg to pattern bghitliyavr smyl ki tyachi ek line आपली print karta yete aani tya aadvya eka line vr 'ni time repeated ☆ ahet aani jr repeted gosht print karaychi asel tr aaplyakde for loop aahe mg for loop use kela pat tich line ajun 3 times repeat zaleli disli pt repeated gosht print karayla for loop use kela. loop la itpasun start kel aani th jevdhi value aahe tith pryant fimt aani increment kart gelo. condition false zalyavr aatla loop thambla bahercha I ne pudhe sarkla

i = 2
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 ← condition false
⇓
inc/dec
condition check
⇓ true
kaam
⇓
inc/dec

☆_☆_☆_☆_☆

~~i = 3~~

i = 3
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 ← condition false
⇓
inc/dec
condition check
⇓ true
kaam
⇓
inc/dec

☆_☆_☆_☆_☆_

i = 4
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 ← condition false
⇓
inc/dec
condition check
⇓ true
kaam
⇓
inc/dec

☆_☆_☆_☆_☆_

i = 5
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 ← condition false
⇓
inc/dec
condition check
⇓ true
kaam
⇓
inc/dec

☆_☆_☆_☆_☆_

i = 6
condition check
⇓ false
kaam

Q2 Print the following Pattern

```
1  1  1  1
2  2  2  2
3  3  3  3
4  4  4  4
```

```
import java.util.*;
class OneTwoThreePattern {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt ();
        for(int i = 1; i<=n; i++){
            for(int j= 1; j<=n; j++){
                System.out.print(i+" ");
            }
            System.out.println ();
        }
    }
}
```

Dry Run

i mT print kely

n = 4

i = 1

condition check

⇓ true

kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 ← condition false

⇓

inc/dec

condition check

⇓ true

kaam

⇓

inc/dec

1_ 1_ 1_ 1_

i = 2

condition check

⇓ true

kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 ← condition false

⇓

inc/dec

condition check

⇓ true

kaam ⟹ 2_2_2_2_

⇓

inc/dec

i = 3
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 ← condition false
⇓ condition check
inc/dec ⇓ true
kaam ⟹ 3_3_3_3_
⇓
inc/dec

i = 4
condition check
⇓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 → condition false
⇓ condition check
inc/dec ⇓ true
kaam ⟹ 4_4_4_4_
⇓
inc/dec

i = 5
condition check
⇓ false.
kaam

Pattern kde bghitlyavr smjl ki first row 1, second row 2, third row madhe 3 .... as disty tyala break kel pahilanda thought generate kela ki 1 chi row kashi print karu shakto. tya row madhe 1 repeted aahe mhantlyavr for loop use karava lagnar he fix zal. thought generate zale ki aaplyala 1 chi ek row print karaychi aahe tr aaplyala print karatana jo outer for loop aahe to print karava lagnar pn to print karatana, tyala initialize. 1 kel aani condition dili ki jo pryant n kiva n peksha motha number yet nahi to pryant tu print kr aani condition true zali ki increment he 1 ne te aaplyala n time pahije mhantlyavr aapn j cha pn loop n times firnar. jevdhe column tevdhe row pahijet tyasathi. j chya loop la pn same condition dili ki initialize 1 pasun ho aani n kiva n peksha motha number yet nahi to pryant first raha aani condition true asel tr ek ne increment ho j ची condition false honar aani i ek position pudhe janar.

Q3 Print the following Pattern
for n = 4

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

```
import java.util.*;
class OneOnePattern {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt ();
        for (int i = 1; i<=n; i++) {
            for (int j = 1; j<=n; j++) {
                System.out.print (j + " ");
            }
            System.out.println ( );
        }
    }
}
```

print j la kely

Dry Run
n = 4
i = 1
condition check
↓ true
kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 → condition false
↓
inc/dec

condition check
↓
kaam ⟹ 1_ 2_ 3_ 4_
↓
inc/dec

i = 2
condition check
⇓ true
Kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 → condition check false
⇓ inc/dec
condition check
⇓ true
Kaam ⟹ 1_2_3_4_
⇓ inc/dec

i = 3
condition check
⇓ true
Kaam ⟹ j = ~~1~~, ~~2~~ ~~3~~ ~~4~~ 5 → condition false
⇓ inc/dec
condition check
⇓ true
Kaam ⟹ 1_2_3_4_
⇓ inc/dec

i = 4
Condition check
⇓ true
Kaam ⟹ j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 → condition false
⇓ Inc/dec
Condition check
⇓ true
Kaam ⟹ 1_2_3_4_
⇓ inc/dec

i = 5
condition check
⇓ false
Kaam

pattern bghitlyavr smjl ki aapli ji 1st row aahe ti 1 2 3 4 n number
pryant aahe. pahila focus tevhla ki to 1 2 3 4 he aapn kas
print karu shakto pr thought generate zala ki jo aapla
inner loop firnar tya loop chi position aahe 1 2 3 4
1234 repeted hoty repeted hoty mhantlyavr for loop use karayla
lagnar pahila for loop ghetla. I la initialize kela n tiva n peksha
motha number yei pryant tyala firayla sangitl condition true
asel tr increment kr aani condition true asel tr pr aat yet

ye aani aatla for loop la prt. initialize kel lla tyala pn same n kiva n peksha motha number yei pryant firayla sangitl aani condition true asel tr aatlya for loop cha position number print kr aani increment kr jevha condition false hoel tevha for loop chya baher pd aani baher cha for loop lne increment kr aani prt procedure chalu rahude